



Module 5

Training Benchmarks

Various methods are used to benchmark a runner's physical condition and performance. These methods are often correlated to accurately assess a client's level of conditioning.

The areas discussed in this module are:

- Rate of Perceived Exertion
- Heart Rate
- VO₂ Max
- Anaerobic Threshold
- Lactate Threshold

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Rate of perceived exertion
 - Perceived intensity assessment
- Use of heart rate in establishing training intensity zones and recovery
- Cardiac drift
- Utilization of rest and recovery heart rate in the training process
- Limitation of VO₂ max
- Exercise economy
- Maximal aerobic heart rate assessment
- Is anaerobic threshold real?
- Assessments of lactate threshold (talk test/MLSS)
- Creation of MLSS and maximum aerobic-based training zones

❖ RATE OF PERCEIVED EXERTION (RPE)

A client's rate of perceived exertion is the measure of how physically and mentally difficult an individual feels an exercise or activity is.

0	No Discomfort
1	Barely Noticeable Discomfort
2	Very Mild Discomfort
3	Moderate Discomfort
4	
5	Strong Discomfort
6	
7	Very Strong Discomfort
8	
9	
10	Unbearable Discomfort

Figure 5.1 Rate of Perceived Exertion Scale (Borg CR10)

The **rate of perceived exertion (RPE) scale** is a subjective method to assess one's physical exertion or pain level. This scale was created in the 1960s by Dr. Gunnar Borg, a Swedish psychologist. Dr. Borg created two RPE scales, the 6–20 scale and the CR10 scale (79). The 6–20 scale is based on a scale from 6 to 20, with 6 representing the lowest exertion/pain level and 20 being the highest level. The CR10 scale is similar to the 6–20 scale. However, it ranges from 0–10 with 0 being no exertion and 10 being the highest. The RPE scale is used in both sports and medical settings.

The UESCA advises its coaches to use the Borg CR10 scale, as the 0–10 scale is typically perceived as more intuitive than the 6–20 scale. Figure 5.1 shows a modified Borg CR10 RPE Scale.

Regardless of the type of RPE scale used, the scale is based on the client's perception of exertion or pain and therefore is highly subjective. For example, someone with a high pain tolerance might rate running at 95 percent of his or her lactate threshold as a 6 on the RPE scale (Borg CR10), whereas someone with a lower pain tolerance might rate that effort level as a 9. While you cannot take the subjectivity out of the RPE scale, you can make an effort to try to assess your client's intensity tolerance. A good way to assess a client's intensity tolerance is by using UESCA's **perceived intensity assessment (PIA)**.

This is an important assessment because it gives you an indication of what your client's interpretation of "intense" is. While you will never be able to say with 100 percent certainty if the RPE equates to the true corresponding physiological level of intensity, the PIA allows you to have a better idea of how physiologically accurate a client's RPE number is.

PERCEIVED INTENSITY ASSESSMENT (PIA)



Figure 5.2 Wall Sit

The PIA uses a wall sit to assess perceived intensity. Wall sits are performed by having your clients “sit” with their back against a wall with their hips and knees at 90-degree angles. Instruct them to keep all their weight on their heels and to lift up the balls of their feet. By performing this modification, most all of the muscular stress is placed on the quadriceps. DO NOT inform clients of the real reason for this assessment. Rather inform them that you are assessing their quadriceps strength.

After instructing them on the correct form, tell them to hold it as long as they can, otherwise known as *holding until failure*. Inform them that they will feel their quadriceps start to burn. However, they should try to push through the initial burning sensation and hold the wall sit for as long as possible. Since this assessment is correlated with the RPE scale, instruct clients that on a scale from 1 to 10 for intensity (10 being the most intense), they should try to hold it until a level 10.

While they are performing the assessment you will want to assess the following areas:

- Visible facial signs of fatigue and discomfort (e.g., grimacing, focused eyes, clenched teeth)
- Visible body signs of fatigue and discomfort (e.g., legs shaking, toes wanting to lower, body starting to lower on the wall)
- After the exercise ends notice if they have to catch their breath or collapse on the floor, or do they casually stand up and ask what is next?

For this assessment, you might find it helpful to assign perceived intensity levels associated with the client's body response. For example, if a client collapses on the ground when his or her legs give out, this could represent a 9 or 10 (out of a 1–10 scale). However, if a client is talking throughout the test and shows no strain when stopping the wall sit, this might be representative of a 2 or 3.

The reason for using this exercise for the assessment is twofold. First, the wall sit is a safe exercise as it is isometric and does not put stress on the lower back. Second, because of the isolation of muscular activity (quadriceps), the muscles fatigue quickly, and therefore it is a relatively quick assessment. Note, while this is a safe exercise, your clients will likely experience muscular soreness in their quadriceps in the following days. This is especially the case if a client is new to resistance training.

In actuality, this assessment is as much a mental assessment as it is a physical one. You are using a physical exercise to perform a mental assessment of your clients regarding their tolerance of exertion. In this case, the mental assessment is based on pain tolerance.

Typically, former or current athletes rate quite high on this test as they are accustomed to exercising at high intensity and have a relatively high pain tolerance. After performing the assessment, you will have a fairly good idea of where your client's intensity and pain tolerance lies. The purpose of this test is not to see how long your client can hold a wall sit for, but rather the physical characteristics of your client throughout the assessment, which in turn, you will use to decipher the individual's level of perceived exertion.

This assessment should be done at the onset of working with a client as well as throughout the training process to assess progression.

❖ HEART RATE

Heart rate training has long been the standard when analyzing and assessing cardiovascular effort. Use of a heart rate monitor (HRM) is a relatively inexpensive and beneficial way to assess effort level. Most heart rate monitors consist of two pieces, a watch and a chest strap. Some activity trackers pick up heart rate at the wrist. Benefits of using an HRM are:

1. Athletes can tell when they are not recovered. Without an HRM, this is hard to determine and forces athletes to rely on how they feel, which is not always the best measure. Below are three metrics that typically correlate to not being recovered and rested:
 - a. Heart rate is slow to return to baseline after an effort.
 - b. Abnormally high/low resting heart rate. An abnormally low heart rate is most common.
 - c. Heart rate does not reach normal peak levels when putting out a corresponding effort.
2. It is especially useful during preseason training when athletes are working on their base fitness level. By using an HRM, athletes can monitor their heart rate throughout the exercise session to make sure they do not push too hard. This is very important because as a general rule of thumb, the more deconditioned athletes are, the higher their heart rate will be and the faster their heart rate will elevate with increased effort. This elevation in heart rate does not always correspond to the athlete feeling tired or putting forth excessive exertion. If you do not use an HRM with clients to measure heart rate during base fitness training, it is very easy for them to exercise too hard and not get the most out of the training process.
3. Training with an HRM allows an athlete to set heart rate training zones. By setting these zones, an athlete can focus on varying intensities with accuracy.
4. Determine running pace.
5. Think of a heart rate monitor as a tachometer for the body. A tachometer in a car measures the number of revolutions per minute of the crankshaft in the engine. In this way, the tachometer shows how hard the car is working. On most tachometers you will see a red zone from approximately 7K RPMs and above.

In this example, the 7K represents the *red line*. The red line is the beginning of an area that produces such high engine RPMs it is dangerous to keep increasing them. This is no different from your heart in terms of the red line. The red line, in human terms, is the *maximum lactate steady state* (MLSS). If athletes were to continue to push past their MLSS, they would be performing on borrowed time unless they lower their

intensity. Like a car, an athlete cannot stay above the red line too long or at too high a level before performance decreases markedly and rapidly.

ESTIMATED MAX HEART RATE

A long-established standard for determining an individual's maximum heart rate has been the formula:

$$220 - \text{Age} = \text{Maximum Heart Rate}$$

Based on the result of this formula, heart rate zones could be set to determine varying intensity workloads. There is just one problem with this formula ... it's a formula! A formula is not practical in determining an individual's maximum heart rate, as there is too much of a range as to what one's maximum heart rate could be. This needs to be assessed on an individual basis, not based on a set formula. Regardless, there is no scientific validity to the formula.

The 220–Age formula was developed in the late 1960s by cardiologist Sam Fox and physiologist William Haskell (350). The formula was derived in an attempt to estimate maximum heart rate for individuals undergoing stress testing. The formula was developed using a small sample of individuals. Interestingly, even Haskell noted that the formula was not intended for use in determining maximum heart rates of athletes (349).

Since this formula was developed, there have been attempts to identify a more accurate and scientific way to predict and estimate maximum heart rate. Perhaps the most valid was a 2001 study by Dr. Douglas Seals, a physiologist. Unlike Haskell and Fox, Dr. Seal's research drew on a data sample of a large number of subjects (over 18K) across hundreds of studies, including his own. What Dr. Seals found in regard to the standard formula (220–Age) is the following (351):

- Overestimates maximum heart rate for young adults
- Fairly accurate for individuals around the age of 40
- Underestimates maximum heart rate in older adults

Based on his research, Dr. Seals proposed a revised formula:

$$208 - 0.7 \times \text{Age} = \text{Maximum Heart Rate}$$

A 2007 study by Gellish et al. (352) corroborated Dr. Seal's results and came up with a near exact formula ($207 - 0.7 \times \text{Age} = \text{MaxHR}$).

Conclusion

It is not advised to estimate maximum heart rate using a formula for the following reasons:

- You are not working in a clinical setting with cardiac patients who require you to determine their estimated maximum heart rates.
- You are working one-on-one with clients, and therefore, can perform more valid and specific assessments on which to base intensity prescriptions.
- Any formula will likely have a relatively large degree of percent error.
- Utilizing MLSS as a benchmark for determining intensity zones is more accurate and functional than utilizing a percentage of maximum heart rate.

Therefore, while a formula is not suggested, this information is critical to know and understand, as the $220 - \text{Age}$ formula is still widely used as an established benchmark for athletic cardiovascular programming.

Cardiac Drift (CD)

This is a phenomenon that relates to a natural increase or upward “drift” in heart rate while the overall intensity (e.g., respiration rate, effort level, caloric burn) remains the same (359). There are two primary reasons for *cardiac drift* (359):

1. Increase in body temperature raises heart rate
2. Decrease in muscular efficiency due to fatigue

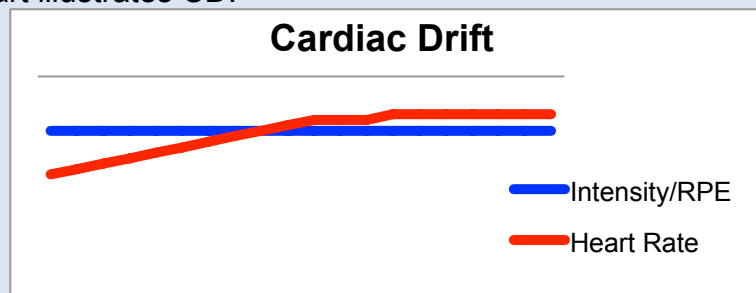
CD has implications when designing a heart rate-based training program. If a client experiences cardiac drift during training sessions that are based around heart rate, there is a strong likelihood that the individual will be underperforming during the training sessions.

For example, let us say a client is supposed to train between a heart rate of 155 and 165 beats per minute (BPM). For the first half hour the client is able to run at this pace easily while maintaining a 7:30 min/mile pace. However, at approximately 35 minutes into the run, cardiac drift brings the heart rate up to 168. At this point, the client slows down to stay in the appropriate heart rate training zone. By decreasing the effort level to accommodate for CD, the client is reducing his or her workload, whereas if the person stayed at the same effort level (even though the heart rate would rise past the prescribed heart rate zone) he or she would be performing at the appropriate effort level.

Therefore, it is important to inform the client about CD and to judge effort not just on heart rate but also on perceived exertion. If you are aware that your client experiences CD as well as how many heartbeats the individual drifts upward, you may consider assigning the client a higher heart rate zone to factor in CD.

In a 2000 study of 10 highly conditioned cyclists performing an MLSS test, CD was noted from the beginning to the end of the test (140).

The following chart illustrates CD:



RESTING HEART RATE

This assessment is performed by taking one's heart rate upon waking and before getting out of bed. This gives a good base measurement in terms of your client becoming more aerobically fit as well as when the person is recovered. Generally speaking, the lower the resting heart rate, the more aerobically fit someone is. Individuals naturally have varying resting heart rate levels, so the value of this test is not the initial resting heart rate but rather the reassessments. Resting heart rate is affected by many things, so to obtain the most accurate baseline measurement, your client will want to take his or her resting heart rate on three or four consecutive days and average the results. Throughout the training process, and especially throughout the base training phase, your client should see the average resting heart rate decline.

A decrease in resting heart rate corresponds to an increase in stroke volume (amount of blood ejected with each heart stroke). When stroke volume increases, the heart beats less to supply the same amount of blood to the body as it would at a lower stroke volume. This physiological adaptation is indicative of the cardiovascular system becoming more efficient and the heart not having to work as hard to supply the body with blood. Typically the greater one's age is the lower the person's resting heart rate will be.

A significantly low or high resting heart rate can be indicative of fatigue.

RECOVERY HEART RATE

In addition to the resting heart rate test, the recovery heart rate assessment measures cardiovascular fitness and, more important, improvement. A client's recovery heart rate is assessed by having the individual get his or her heart rate up to a predetermined level, then ceasing activity. The recovery heart rate is represented by the number of beats the heart rate drops in one minute after cessation of exercise. The greater the heart rate drop, the more aerobically conditioned an athlete is. The speed at which the heart rate drops can also be attributed to the fatigue level of an athlete. If an athlete is not recovered from a previous workout or is sick, the heart rate will not drop as fast as it would otherwise. Therefore, to get a true representation of the correct heart rate drop, the same test should be done on different days and the average calculated. As

your client becomes more aerobically fit, the person's average heart rate recovery time should decrease.

HEART RATE TRAINING ZONES

Results from the cardiovascular assessments will enable you to create intensity-based protocols for your client.

Heart rate training zones allow a client to work within set parameters regarding intensity during training sessions. The higher the training zone number, the higher the intensity level. There are two primary types of heart rate charts:

1. Based on an individual's maximum heart rate and the zones expressed as percentages of the max heart rate. This method is not recommended.
2. Based on an individual's lactate threshold or, more specifically, the person's maximum lactate steady state (MLSS). Using MLSS-based training zones, heart rate levels are integrated. For example, an individual's MLSS Zone #4 might equate to a heart rate zone of 145–150 beats per minute (BPM).

❖ VO₂ MAX

The term ***maximum effort heart rate*** or ***VO₂ max*** is used a lot among endurance athletes and relates to maximal oxygen consumption or, more specifically, an individual's maximal capacity to transport and use oxygen during exercise. The V stands for volume, O₂ stands for oxygen, and *max* refers to maximum. VO₂ max also commonly refers to the test to determine a subject's maximal oxygen consumption. So what does VO₂ really represent? A VO₂ max test can be done on any apparatus (e.g., bike, treadmill, rower, etc.); however, the general method of testing is the same. A subject's oxygen consumption is measured while the intensity of the effort increases. As the intensity of the effort is increased, oxygen consumption also increases, initially as a linear relationship. VO₂ max is represented when a subject's oxygen consumption plateaus, even if the subject is able to continue increasing effort. It is important to note that when performing a VO₂ max test, some individuals are not able to determine their VO₂ max because they fatigue before their oxygen consumption hits a plateau.

Additionally, while it is normal to be able to increase one's VO_2 max through exercise, the absolute limit of one's potential VO_2 max is largely determined by genetics. As noted below, some individuals do not reach a plateau.

According to the *American College of Sports Medicine* (ACSM), an individual's VO_2 max declines 5-15 percent per decade past the age of 30 (45). The good news is that older adults respond to aerobic exercise equally as well as younger adults in terms of increasing VO_2 max. This means that older adults can increase their VO_2 max by 10-30 percent through aerobic exercise (45).

In regard to increasing one's VO_2 max, studies have found that increases in VO_2 are mostly limited to recreational athletes versus well-trained (elite) athletes (597, 598). Therefore performance increases in well-trained athletes are likely due to increases in running economy versus increases in VO_2 max.

An individual's VO_2 max is typically expressed as milliliters of oxygen per kilogram of body weight per minute (ml/kg/min). For individuals who either should not or cannot do the VO_2 max test, there are estimated tests that do not take the individual to his or her absolute maximum. These tests are also called *submaximal tests* and use equations to estimate VO_2 max. While the VO_2 test determines aerobic fitness, this test in isolation has often been found to be a poor determining factor of success in endurance athletes as there are many other factors that influence athletic performance (46). Therefore, determining one's VO_2 max is a great way to benchmark fitness and use for reassessment purposes, but in regard to performance in the real world, it is only one potential factor.

EXERCISE ECONOMY

The primary reason why VO_2 max is often viewed as a poor indicator of performance is because of *exercise economy*. Exercise economy relates to how efficient an athlete is. Many factors influence one's exercise economy, such as aerobic fitness, biomechanics, and muscular endurance. To demonstrate how exercise economy fits into overall performance, let's look at two hypothetical athletes. Both athletes (athlete A and athlete B) have a VO_2 max of 70. However, athlete B is much more biomechanically efficient than athlete A. If one were to look at just VO_2 max, they would be identical in regard to potential. However, when you look at the big picture (exercise economy), it becomes clear that

athlete B will outperform athlete A. In other words, athlete B has greater exercise economy than athlete A.

When assessing clients in regard to their potential as well as their current level, you must take exercise economy into account.

To drive home this point, take for example Paula Radcliffe, holder of the women's marathon record (2:15:25). When tested (VO_2 max) initially as an 18-year-old, and 11 years later, her VO_2 max did not change despite her vast improvement. Therefore it is easy to see in this case that her improvement was largely due to an increase in running economy, not an increase in VO_2 max (494).

Sport-Specific Fitness?

How is it possible that an individual could run a 2:30 marathon but barely make it one lap in a pool without hyperventilating and experiencing a skyrocketing heart rate? Is the person out of shape? Not likely if the individual can run a 2:30 marathon. They are, however, inefficient when it comes to swimming, and the wasted energy equates to rapid rate of fatigue and elevated heart rate. This individual's performance in the pool has no correlation whatsoever to aerobic fitness level. It is just a question of what sport the person is most efficient at.

While cardiovascular fitness is not sport specific, an individual's overall efficiency is, which directly impacts the demand on the cardiovascular system.

Running Economy

While many variables influence running economy, leg girth and, more specifically, girth of the lower leg greatly affect the efficiency of a runner. The heavier the leg, the more energy it takes to swing it. The closer to the foot the weight is, the truer this becomes. For example, a runner would have greater exercise economy if he or she had large quadriceps and small calves versus large calves and small quadriceps.

A study by Myers et al. looked to replicate this scenario using weights applied to different parts of the body while running (492). When 8 pounds were placed at waist level, it required 4 percent more energy than when running without any extra weight. However, when a 4-pound weight was attached to each ankle, it required the runner to exert 24 percent more energy to run the same pace.

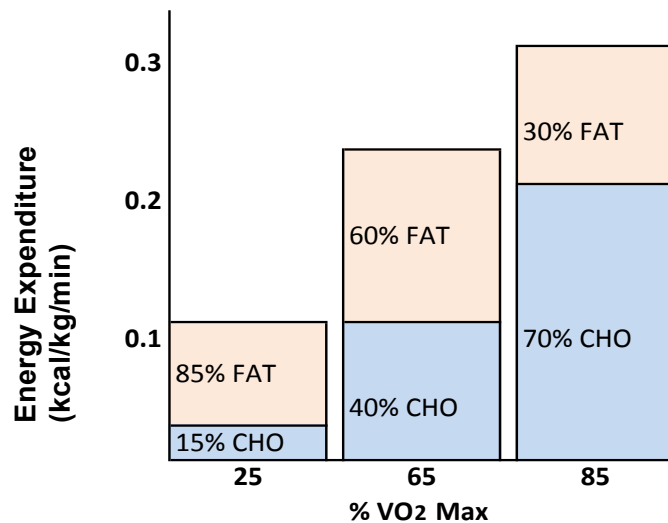
Another study compared two groups of boys from different countries. The average volume/thickness of one group's calves was 15–17 percent less than the other group. When running, this equates to energy savings of 8 percent per kilometer (493)!

Relating VO₂ Max to 'Burning Fat/Calories'

If you were to ask 10 people what the correct workout intensity is to lose weight, burn calories, and “get in shape,” you would most likely get 10 different answers. One of the reasons for this is because you are really asking three different questions.

Many people associate a great workout with high intensity. However, given the physiological response to aerobic training, this is not necessarily the case and is largely dependent on a person's fitness goals. Most cardiovascular machines (e.g., treadmills) have a chart that depicts heart rate zones that are optimal for burning fat or calories and getting into aerobic shape. It is true that the lower the intensity at which an individual trains, the greater the percentage of energy comes from fat. However, one needs to look at it from the perspective of total energy expended. This was demonstrated by Johannes Romijn et al. in a 1993 research paper titled *Regulation of endogenous fat and carbohydrate metabolism in relation to exercise intensity and duration* (47).

Romijn demonstrated that while at 25 percent of an individual's VO₂ max, the percent energy from fat was 85 percent. At 65 percent of an individual's VO₂ max, the percent energy from fat decreased to 60 percent. However, due to the increased energy expenditure at 65 percent of VO₂ max, the 60 percent of energy from fat accounted for more (about 40 percent more!) than the energy from fat at 25 percent VO₂ max. An individual at 85 percent of his or her VO₂ max uses only 30 percent energy from fat. Even though this percentage is low, it still represents more total energy from fat than someone at 25 percent VO₂ max, but less than someone at 65 percent VO₂ max (47). The following chart graphically illustrates this.



* CHO = Carbohydrate

Figure 5.3 Benchmarking VO₂ max in relation to burning fat and carbohydrates

Therefore, when discussing which intensity is more optimal for burning fat, there are two answers: at 25 percent VO₂ max, you will have the greatest percentage of energy coming from fat. However, the intensity where you will burn the most fat is at approximately 65 percent VO₂ max. To address the calorie issue, this one is pretty easy. The greater the intensity, the greater the caloric expenditure will be.

This chart is also a graphic representation of how exercise intensity correlates with exercise sustainability. The greater the intensity, the greater the reliance upon carbohydrates, and therefore, the less sustainable the exercise is for long durations of time.

Why Not Assess VO₂ Max?

The VO₂ max test is considered the gold standard in determining aerobic capacity. This test requires specialized equipment and training to administer. Submaximal tests are used to estimate the VO₂ max for individuals for whom a true VO₂ test would be unsafe or for those who do not have access to a VO₂ max test, or simply do not want the full test. As a side note, a VO₂ max test typically costs \$100–\$300 and is often done in conjunction with a blood lactate threshold test.

As noted previously, VO₂ max is often a poor indicator of performance, as it does not take into account exercise economy. This is where correlating blood lactate is

valuable to know. For example, two athletes could have the exact same VO_2 max but different blood lactate levels. VO_2 max relates to aerobic capacity, but lactate threshold (LT) determines what percentage of their VO_2 they can perform at and for how long. This was demonstrated in a 1982 study by Bertil Sjodin (499). In this study, runners performed training sessions at lactate threshold for 14 weeks. At the end of the 14 weeks, the subjects decreased their average mile time from 5:43/mile to 5:29/mile. Interestingly, the VO_2 max of the runners did not increase at all!

A 2008 study by Tim Noakes found that during a VO_2 max test, only 47 percent of subjects reached a plateau (728). While this finding suggests that the limit of a VO_2 max test may not be solely dependent on oxygen, it is clear that the defining point of a VO_2 max test (plateau) is likely not reliable.

Quite a few assessments/tests are used to estimate (submaximal) VO_2 max. The *Bruce protocol test* (70) and the *YMCA cycle ergometer test* are two of the more widely used assessments. In relation to a true VO_2 max assessment, estimated VO_2 max assessments are relatively easy to administer and do not require extensive equipment or training.

While it might be interesting to know one's VO_2 max, from a training perspective, you should be more focused on the heart rate – specifically the final heart rate when the client can no longer continue. This is because VO_2 max expressed as ml/kg/min cannot be correlated directly to other aerobic fitness benchmarks, whereas heart rate can.

When compared to a true VO_2 max test, many of the estimated VO_2 max tests underestimated one's VO_2 max (454, 455). Specifically, the *Bruce protocol test* and the *YMCA cycle ergometer test* were shown to underestimate VO_2 max by an average of 14 percent (455). For the preceding reasons, this certification will not include an estimated VO_2 max assessment.

ASSESSING MAXIMAL AEROBIC HEART RATE

While one's VO_2 max expressed as ml/kg/min does not hold substantial value in relation to benchmarking training parameters, one's maximal aerobic heart rate is of value for the creation of training zones. The protocol to assess maximal aerobic heart rate is noted below.

The *maximum aerobic heart rate* will be used to help benchmark LT-based (MLSS) training zones.

Prior to the end of the assessment, the increase in heart rate will typically slow or plateau. This is indicative of a maximum or near-maximum effort.

Variables with the Maximal Aerobic Heart Rate Assessment

1. Clients who are extremely deconditioned will likely not be able to reach their maximum aerobic heart rate level due to systemic fatigue.
2. If clients have a low pain tolerance, they may stop the test before their maximum aerobic heart rate is reached.
3. If your clients are new to running, their heart rate may be “artificially” high due to a low running economy. Additionally, their legs may fatigue before they reach their aerobic capacity.
4. If your clients are fatigued, their heart rate may be higher or lower than normal, depending on how their body responds to stress when fatigued. Typically the heart rate response is reduced.

If you believe that your client fits into any of these scenarios, be advised that the maximum aerobic heart rate may have some degree of error.

Following are two protocols used to establish the **maximal aerobic heart rate**.

Treadmill

This is a modified assessment based on protocol from Persinger et al. (2004) (107).

1. Warm up at an easy walk for 15 minutes.
2. Set the treadmill at a speed at which the client feels he or she could walk or jog indefinitely. At this point, the starting gradient is zero.
3. Every two minutes, the incline is increased by 2 percent.
4. Denote the client's heart rate at one-minute intervals.
5. The test should continue until the client can no longer continue. At this point, note the final heart rate. This represents the client's maximal aerobic heart rate.

The following protocol is more specific to running, as it places an emphasis on an increase in speed versus an increase in incline.

1. Warm up at an easy walk for 15 minutes.
2. Set the treadmill at a speed at which the client feels he or she could walk or run indefinitely. The starting gradient is .5 degrees incline.
3. Every two minutes, increase the speed one mile per hour.
4. Denote the client's heart rate at one-minute intervals.
5. The test should continue until the client can no longer continue. At this point, note the final heart rate. This represents the client's maximal aerobic heart rate.

Since these assessments rely on controlled, incremental speed/incline increases, a treadmill is required.

As the intensity is quite high for this assessment, your client must get a physician's clearance prior to participating in the assessment.

❖ ANAEROBIC THRESHOLD: FACT OR FICTION

The term **anaerobic threshold (AnT)** is used extensively in the world of endurance sports and most commonly used to describe the working upper limit of one's aerobic capacity. AnT is difficult to define as it is more of a concept than an established metric. It should be noted that some individuals do not believe that AnT exists (131). Because of this, there are many different interpretations regarding what AnT is. In addition to confusion about what the definition of AnT is, it makes sense that there is equal confusion over how to assess and estimate it (441). Some of the most common benchmarks used to interpret AnT are:

- **Lactate Threshold (LT)**
 - Representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.
- **Maximum Lactate Steady State (MLSS)**
 - The highest intensity level at which blood lactate concentrations are maintained at a steady-state (equilibrium) level during exercise bouts of at least 60 minutes (441). This may or may not be higher than OBLA.
- **Ventilatory Threshold (VT)**
 - The point at which the ventilation rate increases faster than the workload. Until the VT is reached, the workload and respiration rate increase linearly. Often correlated with LT.
- **Onset of Blood Lactate Accumulation (OBLA)**
 - OBLA is often used synonymously with LT. However, OBLA occurs at a slightly higher level than LT (71, 125). Technically speaking, OBLA is reached when the blood lactate accumulates to a certain level in the blood (4mmol) (382).

A primary reason for the increase in popularity of quantifying AnT in sports is the poor correlation between VO_2 max and athletic performance (137, 138). As a result, finding a submaximal threshold that can be maintained over long periods has gained traction within the endurance sports community (138).

While the AnT interpretations just noted have value to runners when gauging intensity, they should not be used to directly infer AnT (127). This is primarily because of a lack of correlation between assessment methods. For example, a 1993 study out of the University of British Columbia found that when using VT and LT to estimate AnT, many of the correlations appeared to be coincidental (123).

The theory that a coincidental correlation exists between AnT, VT, and LT is further supported by a 1982 study that looked at hyperventilation rates in people with *glucose storage disease-type 5* (commonly called *McArdle syndrome*) (141, 142). Individuals with McArdle syndrome cannot break down glycogen in the muscles. As blood lactate is the result of glycogen breakdown (catabolism), these individuals do not produce lactate above resting levels with exercise. In this study, the participants' VT increases at the point at which their proposed AnT would be. This demonstrates that VT is not caused by an increase in LT.

Therefore, the assessments that are noted above, as well as others, should be used as stand-alone assessments, but not for the purposes of inferring AnT.

The implication of a proposed AnT is that once individuals exceed it, they are no longer using oxygen for energy but rather relying on glycogen exclusively for energy. This is erroneous as there is no evidence that the body “switches” from using oxygen to not using oxygen at a particular intensity level. Oxygen is present at all levels of intensity, albeit in varying amounts. At one's proposed AnT, the anaerobic energy systems do not take over from the aerobic energy system, but supplements it to meet the increased energy demand (126).

Origin of Anaerobic Threshold

The term *anaerobic threshold* was coined by Wasserman and McIlroy in 1964 as a way to identify an intensity that was challenging and safe for cardiac patients (136). The goal of anaerobic threshold was to identify a reliable, submaximal effort level so that cardiac patients were not subjected to maximal cardiovascular tests. The premise of their anaerobic threshold theory was that blood lactate accumulation is the result of poor oxygen levels (127).

❖ LACTATE THRESHOLD (LT)

Lactate threshold is representative of the level at which blood lactate accumulates in the bloodstream. This occurs when lactate production exceeds lactate clearing.

While there is always lactate being produced, at rest or at low to moderate intensities of exercise, the lactate is “cleared” within the muscle. So what exactly does “cleared” mean and how does this influence LT? In regard to LT, when you see the term “cleared,” think “used for energy.” Lactate is cleared intra-muscularly by (319):

- Oxidizing the lactate and converting it back to pyruvate (oxidative fuel) within the muscle in which lactate was formed.
 - Lactate is used for fuel in the mitochondria. Lactate permeates the mitochondrial wall via a special transporter protein called *monocarboxylate transporters* (MCTs) (512). MCTs can be increased via training, thus facilitating increased lactate clearance (76).

This clearing method prevents the intra-muscular lactate levels from increasing too fast to the point where lactate levels reach the intra-muscular threshold and spill out into the bloodstream. Therefore intra-muscular lactate levels can increase substantially (up to five times resting levels) without any increase in blood lactate levels (513). However, when the intra-muscular lactate threshold is breached, some lactate exits the muscle and enters the circulatory system and thus increases blood lactate levels. This is representative of the lactate threshold.

An analogy to this would be to equate a muscle fiber to a drinking glass, the water within the glass to lactate, and any area outside the glass to the circulatory system. Levels of water (i.e., lactate) can increase and decrease, but so long as the levels do not breach the top of the glass (i.e., lactate threshold), blood lactate levels will not increase. However, once the water spills over the top of the glass, the blood lactate levels increase as there is an increase in lactate within the circulatory system.

Once blood lactate enters the circulatory system, it is shuttled to areas of the body that can use it for fuel (i.e., brain, heart, other skeletal muscle). Blood lactate that is not utilized is sent to the liver, where it is converted to pyruvate and then to glucose. The glucose is sent (via blood) to the muscles to be used as fuel. This is representative of the *Cori cycle* (515).

When exercising at MLSS, most lactate is oxidized for fuel by converting lactate to pyruvate and back to lactate. When the intensity increases much beyond MLSS, the lactate enters the circulatory system.

The process by which lactate moves intra-muscularly and throughout the circulatory system is termed the *lactate shuttle* (319, 511).

As lactate can be added and removed from the blood, the amount of blood lactate at any given point is representative of blood lactate accumulation, not production.

Historical Reference

In addition to Dr. Meyerhof's frog study (referenced in *Module 3*), in which he concluded that high levels of lactate caused muscular fatigue (24), two other studies have had a profound impact regarding the erroneous perception of lactate.

In 1857, a biochemist named Louis Pasteur deduced that lactic acid formation was possible only in the absence of oxygen (anaerobic). This is called the *Pasteur effect* (509).

Secondly, in 1923, research by physiologist A.V. Hill theorized a correlation between high work levels and high blood lactate levels. More specifically, he theorized that at high intensities, the aerobic system was not adequate to provide fuel to the body and therefore the anaerobic/lactic acid system would "switch on" and take over from the oxidative system (510).

In the years since then, the following has been found to be true:

- Lactic acid does not exist within the human body.
- Lactate is a fuel.
- Lactate is produced at all times.

- The proposed anaerobic system does not “switch on” after the oxidative system is exhausted.
- Lactate is not responsible for muscle burn.

Despite these findings, there is still a large consensus that believes that lactate is a waste product and responsible for a reduction in human athletic performance.

BENCHMARKING LACTATE THRESHOLD

LT has become a common benchmark for gauging intensity during training and racing in endurance sports. Like anaerobic threshold, the method by which LT is assessed and quantified is confusing. LT benchmarks such as OBLA, MLSS, and LT itself are often viewed as synonymous. As noted previously, LT, OBLA, and MLSS occur at slightly different levels of intensity.

Perform an online search of LT assessment methods and you are sure to find quite a few different methods, charts, graphs, and philosophies regarding the best way to assess and benchmark LT. However, for the purpose of this certification, the only benchmark that we are interested in is MLSS. This is because MLSS has the highest correlation to endurance training and racing.

MLSS Associated Heart Rate

Historically, heart rate as a percentage of one’s estimated maximum heart rate was the primary indicator of intensity. Heart rate has a valid place in benchmarking intensity. However, it is best used when correlated with MLSS. The key reason MLSS is the new standard for gauging intensity is due to its direct correlation with muscle fatigue, and thus, sustainability of exercise. When using an accurate MLSS level as a benchmark, an athlete is able to perform at sustainable levels for relatively long durations.

According to running coach Dr. Jason Karp, running at one’s LT should feel “comfortably hard” (748).

Lactate threshold typically corresponds to approximately 85 percent of one’s maximum heart rate (737).

It is the position of UESCA that the more data points there are, the more accurate it is to benchmark intensity and thus training zones.

The two primary data points that UESCA looks to determine are:

- Maximum Aerobic Effort Heart Rate
- Maximum Lactate Steady State (MLSS)

By performing these assessments, you will be able to accurately create LT-based heart rate zones. However, of the two assessments, MLSS is the more useful.

BLOOD LACTATE CLEARANCE

Lactate clearance was discussed earlier in terms of being cleared intra-muscularly. However, once lactate threshold is reached and lactate enters the circulatory system, lactate becomes blood lactate and is cleared in one of two ways (29):

1. Lactate is shuttled to areas of the body that can use it for fuel (i.e., brain, heart, other skeletal muscle).
2. Blood lactate that is not utilized is sent to the liver, where it is first converted to pyruvate, then to glucose. This is representative of the *Cori cycle*.

When blood lactate levels increase past what can be cleared, blood lactate begins to accumulate.

In a 1989 study by McMaster et al., swimmers were assessed to determine the best intensity level to clear lactate after a maximal effort. McMaster et al. found that when swimmers swam at 65 percent of their maximum effort, they cleared the most lactate (320). This study was performed using swimmers because during meets, swimmers often have to compete in multiple events, thus necessitating proper recovery from one event to the next.

Another study found that exercising near to one's LT cleared the most lactate. This study also found that lactate was cleared faster via active rest than at rest (506).

While the first study noted is specific to swimming, the science likely holds true for that of running. It is applicable from both a perspective of recovering during competition as well as during training when performing interval-type sessions.

The positive effects of active recovery in regard to clearing lactate seem to be limited to 20 minutes or less (507).

Training Implication

While physical training has no impact on the body in regard to lactate production, it does influence the ability of an individual to clear lactate. Interval-based training is the most efficient way to increase one's ability to clear blood lactate.

ESTIMATED LT-BASED ASSESSMENTS

Conconi Test

A popular estimated anaerobic threshold (AnT) test is the *Conconi test*. The Conconi test was developed in 1982 by Francesco Conconi, an Italian sports doctor (259). This test can be administered on either a treadmill or bike. This test determines AnT by plotting a subject's heart rate and looking for a plateau. This plateau is known as the *deflection point* and represents the AnT. The theory behind the Conconi test is that one's heart rate increases linearly with the increase in workload, and when the linear relationship ends because of the deflection point it is representative of the AnT.

Individuals can continue exercising past their AnT and their heart rate will continue to rise. However, the individual will probably not be able to continue for much longer. Several studies have challenged the validity of the Conconi test, stating that not all subjects have shown deflection points on the test, meaning the results are not consistently replicated (73). Also, studies show a lack of correlation between the Conconi test and true LT tests. The lack of correlation demonstrates that the Conconi test likely overestimates LT (74). While UESCA will not use the Conconi test for the aforementioned reasons, it is important to be aware of this test in regard to its intended use and the challenges to its validity.

Why Not Directly Test for Lactate Threshold?

LT can be assessed directly by taking samples of blood for analysis and running them through a lactate blood analyzer. However, unless you are trained on how to administer and analyze an LT test, it is recommended that you outsource this test to a qualified professional. If you are qualified to administer the test, you must use all proper blood handling/disposal procedures and must check with local laws to ensure you are complying with current health regulations. For most coaches this is not an option, and therefore, an estimate assessment of LT (MLSS) is the best course of action (72). An estimated LT assessment such as MLSS is an accurate, safe, and legal way to assess an individual's LT.

INTERPRETING LACTATE THRESHOLD VIA TALK TEST

The **talk test** is a basic and simple way to estimate lactate threshold. The talk test has three levels of intensity:

1. Below threshold
2. At threshold
3. Above threshold

The talk test was initially proposed in 1939 by Professor John Greyson of Oxford University as a way of advising mountaineers on how to pace themselves while climbing (250). At the time, this advice was not scientifically based, but rather a general observation. However, more recently, the talk test has been shown in many studies to correlate highly to LT. Interestingly, the test was shown to correlate higher to LT than VT (107, 247, 248, 249).

While the testing protocol varies between studies, the gist is that the intensity of the exercise increases incrementally at set intervals and at the end of each interval, the subject recites a sentence or phrase. Additionally, some testing criteria had subjects state their speaking comfort.

- A. Comfortable (below threshold):** can speak without difficulty
- B. Not sure (at threshold):** client can speak, but speech is slightly labored
- C. Uncomfortable (slightly above to substantially above threshold):** can speak only with extreme difficulty or not at all

The subject recites a phrase as opposed to stating one's speaking comfort level, as this is largely subjective.

The talk test can be done on any apparatus. However, given the specificity to running, the treadmill is recommended. The heart rate should be recorded at the end of each two-minute interval along with the ability level to recite a predetermined phrase, based on the criteria stated above. Below are the protocols for each.

The primary purpose of the talk test in relation to this certification is to establish a starting point (benchmark) for the MLSS assessment. However, if your client is an experienced athlete who knows his or her approximate LT-based heart rate, the talk test is not a prerequisite to the MLSS assessment. If your client is being reassessed for MLSS, the talk test is also likely not a prerequisite.

Treadmill

This assessment is based on protocol from Persinger et al. (2004) (107).

1. Warm up at an easy walk for 15 minutes.
2. Set the treadmill speed at a pace clients feel they could walk or jog indefinitely. The starting gradient is zero.
3. Every two minutes, the incline is increased by 2 percent. During the last 30 seconds of each two-minute stage, a predetermined phrase is recited. For example, in one of the talk test studies, the subjects recited the Pledge of Allegiance. At this point, record the subject's heart rate and ability to recite the phrase given the exertion level.
4. The test should continue until the client has difficulty reciting the phrase, which is indicative of the individual having reached his or her lactate threshold (75).

For a protocol that is more specific to running, instead of increasing the elevation every two minutes, increase the speed one mile per hour every two minutes. Keep the treadmill elevation at .5 degrees throughout the entire assessment.

INTERPRETING LACTATE THRESHOLD VIA MLSS

This test needs to be done at the maximum level (steady state) a client can hold for 30 minutes (76). This typically equates to an intensity level slightly above one's LT.

UESCA advises that the talk test be done prior to the MLSS assessment for two reasons:

1. For the purposes of accuracy, the result of the talk test can be correlated with the result of the MLSS. The talk test will likely be at slightly lower heart rate level than the MLSS.
2. The result of the talk test is used to benchmark the starting point for the MLSS test. The MLSS starting intensity should be slightly higher than the talk test.

Research by Palmer et al., *A 1-day maximal lactate steady-state assessment protocol for trained runners*, (456) demonstrated a strong correlation between the following variables and MLSS.

- Running at 87 percent of one's maximum aerobic heart rate
- 32 breaths per minute (correlates specifically with ventilatory threshold)
- 10K running pace
- 12 out of 20 RPE intensity using the *Borg 6-20 RPE scale*
 - Equates to a level 6/10 on the Borg CR10 scale. Denoted on the Borg CR10 scale as between *discomfort* and *strong discomfort*.

Warm-Up Protocol

The warm-up for the test is a total of 20 minutes. Below is the warm-up structure:

1. 0–5 Minutes: easy, RPE level of 1–3
2. 6–7 Minutes: increase effort to slightly above threshold (based on talk test)
3. 8–10 Minutes: easy recovery period
4. 11–12 Minutes: same effort at step #2
5. 13–20 Minutes: easy recovery period

Test Protocol for 30-Minute Time Trial (Treadmill)

1. Warm up for 20 minutes (based on above warm-up protocol). Keep treadmill at .5 percent incline throughout the test.
2. Gradually build up intensity by increasing the treadmill speed for five minutes so that once the five-minute point is reached, the client is at the LT-based heart rate (based on the talk test) and the test begins.
3. For the next 30 minutes, record the heart rate every minute.
4. Treadmill speed should be slightly higher than the Talk Test when the subject was at the estimated LT.
5. Speed should be adjusted so that the LT effort can be sustained for the duration of the test.
6. Average the heart rate for the last 20 minutes. This number represents the heart rate marker for the client's bottom range of the LT zone (MLSS).

As noted in step #4 of the preceding test protocol, the settings should be slightly higher than the talk test. This is because MLSS occurs at a slightly higher intensity level than LT. A 2011 study demonstrated that MLSS was not reached during the talk test (443).

MLSS Field Test

As noted previously, Palmer et al. discovered a strong correlation between one's 10K running pace and MLSS. Therefore, running a 10K at race pace is advised if you do not have access to a treadmill. However, even if you do have a treadmill at your disposal, running a 10K outdoors is a more “realistic” way to assess MLSS.

The advantage of a treadmill MLSS assessment is that it is a controlled environment, whereas running outside has quite a few variables (i.e., terrain, wind, temperature variations). All affect pace and intensity. Therefore, when performing an MLSS field test, ideally the test should be done on a running track or a flat road with little wind and a temperature range between 55 and 70 degrees. Of course, this is not always possible. This test should be performed only on an individual who is adapted to running from both cardiovascular and musculoskeletal standpoints. Below is the MLSS field test protocol:

Test Protocol for MLSS 10K Time Trial (Outdoor)

1. Warm up via walking or an easy jog for 10–15 minutes
2. Begin 10K effort
 - a. Start HRM recording and stopwatch at beginning of test
3. At end of test, get the HR average and total time (breakout pace per mile)

CREATING MLSS/MAX AEROBIC-BASED TRAINING ZONES

As there are various ways to construct a training intensity chart, the number of intensity zones is up to you. However, it is recommended that you have no fewer than three zones. Generally speaking the more advanced the client is, the more training zones there are. The sample chart below has six training zones.

The chart that UESCA has constructed takes into consideration the results of the *MLSS* and the *maximum aerobic heart rate* assessments and their respective heart rates. The chart has three distinct zones, with each zone further divided into half.

ZONE 1A	Recovery
ZONE 1B	Aerobic Base
ZONE 2A	Aerobic Development
ZONE 2B	Threshold
ZONE 3A	Supra Threshold
ZONE 3B	Max Aerobic

Figure 5.4 MLSS/Max aerobic heart rate-based training zones

In the example below, we assume that the client has the following metrics:

- **Maximum Aerobic Heart Rate:** 182 bpm
- **MLSS:** 157

Below are the steps to construct this chart:

1. Denote the heart rate from the *maximal aerobic heart rate assessment* (182 bpm). This will represent the upper limit of the *max aerobic effort* heart rate zone. Anything past this intensity level is considered *anaerobic*.
2. Half of the maximum aerobic effort heart rate equates to the bottom range of Zone 1A.
 - $182 \div 2 = 91$
3. Denote the heart rate of the MLSS Assessment (157 bpm). This will represent the top range of the threshold zone.
4. For the upper limit of Zone 1B (aerobic base), add the results of steps 2 and 3 and divide by 2.
 - $91 + 157 = 248 \div 2 = 124$

Based on these metrics, the mock-up below denotes the heart rate zone limits.

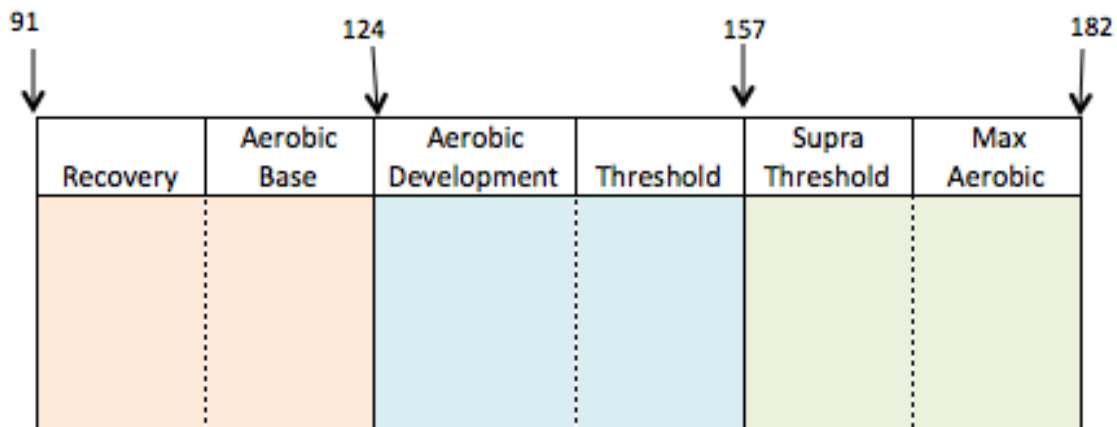


Figure 5.5 Step 1: Creation of MLSS/Maximum aerobic heart rate training zones

To get the interzone heart rate limits, add each of the low/high limits and divide by two. **Example:** Zone 1: $124 + 91 = 215 \div 2 = 108$

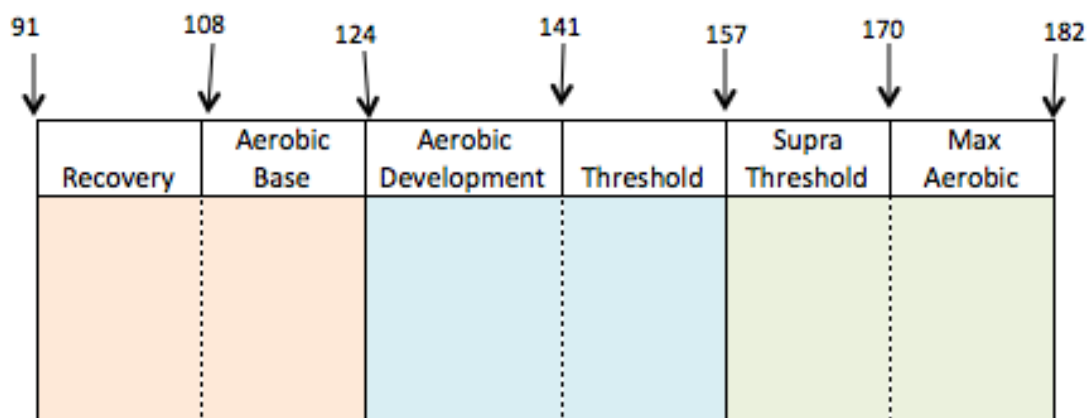


Figure 5.6 Step 2: Creation of MLSS/Maximum aerobic heart rate training zones

Below is what this table would look like for the client:

	ZONE 1A	ZONE 1B	ZONE 2A	ZONE 2B	ZONE 3A	ZONE 3B
	Recovery	Aerobic Base	Aerobic Development	Threshold	Supra Threshold	Max Aerobic
HR	91–108	109–124	125–141	142–157	158–170	171–182
MLSS %	58–68	69–79	80–89	90–100	101–108	109–116
RPE	0 to 2	2 to 3	3 to 4	4 to 5	6 to 7	7 to 9

Figure 5.7 Step 3: Creation of MLSS/Maximum aerobic heart rate training zones

To determine the MLSS percentage zones, you first must determine what the MLSS is. In the preceding example, the MLSS is 157. This is your starting point, and since we are basing the training zones off MLSS, 157 equates to 100 percent of MLSS. To determine the low/high zone percents, divide the MLSS (i.e., 157) into the low/high heart rates for each zone. For example, for Zone 1A, the heart rate zone is 91–108 BPM. To get the low percentage, using a calculator, enter $91 \div 157 = .579 \times 100 = 57.96$ (round up to 58). Use this system to determine the other zones.

The RPE is subjective for each individual. However, the RPEs noted in the preceding example represent rough guidelines for the different training zones.

➤ **ZONE 1A: RECOVERY**

This intensity zone is representative of an active recovery. In regard to intensity, it should feel extremely easy.

- **Training Applications:** shake-out run, spinning with very light resistance on pedals

As noted previously, 65 percent of one's max aerobic heart rate is optimal for clearing lactate. Therefore, if your client is performing interval-based training, based on the aforementioned max heart rate of 182 bpm, the advised heart rate in between interval efforts should be approximately 118 bpm (in the middle of zone 1B).

➤ **ZONE 1B: AEROBIC BASE**

Base training and any type of long workout for the purpose of building or maintaining endurance fall into this category.

- **Training Applications:** LSD, long run

➤ **ZONE 2A: AEROBIC DEVELOPMENT**

This is a slightly more intense version of Zone 1B. During this phase, lactate production increases but is still able to be cleared effectively.

- **Training Application:** fartleks

➤ **ZONE 2B: THRESHOLD**

During this phase, lactate accumulation continues to increase to just below MLSS.

- **Training Application:** tempo

➤ **ZONE 3A: SUPRA THRESHOLD**

At this point, lactate accumulates faster than it can be cleared.

- **Training Applications:** intervals, repeats

➤ **ZONE 3B: MAX AEROBIC EFFORT**

During this zone, an individual is working well above MLSS, and it is not sustainable for long periods.

- **Training Applications:** like Zone 3A, but at a higher intensity such as hill sprints

In addition to stating exact heart rate ranges, you can also direct clients using the zones (i.e., perform 400-meter repeats at the high end of zone 2B). While this does not denote exact heart rate parameters, it offers a client a targeted intensity zone to train in.

Specificity of Sport Disciplines

In relation to the most common endurance sports – running, cycling, and swimming – running typically has the highest heart rate response (at the same effort level, the heart rate is higher), followed by cycling, then swimming. The greater a client's proficiency at a particular sport discipline, the lower the heart rate response will be. This is a function of one's efficiency.

❖ SUMMARY

- A heart rate monitor is an important tool to assess intensity as well as recovery.
 - Using a heart rate monitor to assess one's resting and recovery heart rate helps to assess recovery as well as increases in aerobic fitness level.
- *Rate of perceived exertion* (RPE) scale is a subjective test that measures physical exertion or pain.
 - As the RPE scale is subjective, the *perceived intensity assessment* (PIA) helps determine a subject's exertion tolerance.
- VO_2 max in isolation is often a poor determining factor of performance.
- It is not advised to use one's estimated maximum heart rate to base exercise intensities off of.
- The greatest amount of fat "burn" occurs at approximately 65 percent of VO_2 max.
- The *Conconi test* has been shown to often overestimate lactate threshold.
- The *talk test* can be used in isolation or to establish a benchmark for the *maximum lactate steady state* (MLSS) assessment.
- *Anaerobic threshold* (AnT) is difficult to define as it is more of a concept than an established metric.
- Of all the AnT and LT benchmarks, MLSS has the highest correlation to training for and competing in distance running events and endurance sports as a whole.
- It is advised to create training intensity charts based off the MLSS and maximal aerobic heart rate assessments. However, if using only one assessment to create a chart, the MLSS is the advised assessment to reference.
- *Cardiac drift* relates to a natural increase or upward "drift" in heart rate while the overall intensity remains the same.